

Rudraksh Nanavaty

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Education

Masters	Computer Science , North Carolina State University	2026 - Present
Bachelors	Computer Engineering , Pandit Deendayal Energy University (India)	2021 - 2025

Experience

FirstPeak.ai, Software Engineer Sept 2025 – Feb 2026

- Built Python-based backend services for a multi-channel conversational AI platform; optimized async workflows to handle 50k+ concurrent requests with minimal latency.
- Implemented end-to-end usage tracking and billing, integrating Stripe subscriptions/checkout + metering to enable paid plans and automated invoicing/reconciliation workflows.
- Designed an agent-to-agent evaluation framework where judge agents orchestrate calls to other agents, reducing manual QA cycles by 70% and speeding up releases.
- Parallelized automated voice-agent testing by handling multiple concurrent async Twilio calls, cutting regression suite runtime from 4 hours to 25 minutes.
- Integrated Langfuse observability (trace + latency/cost visibility) to debug slow runs and optimize prompts/model routing.

New Engen, Software Engineer Mar 2024 - May 2025

- Migrated a JavaScript monolith into a TypeScript + Python microservices architecture, improving deployment frequency by 40% and reducing runtime errors by 25%.
- Built Python GraphQL APIs to power custom client dashboards, sustaining 100ms average response times even during 3x traffic spikes.
- Implemented a fault-tolerant bulk emailing pipeline using RabbitMQ, ensuring 99.99% delivery success across server restarts and network partitions.
- Introduced idempotency + durable job processing for email sends (e.g., dedupe keys, retries, dead-letter handling), preventing duplicate sends and guaranteeing at-least-once processing semantics.
- Established performance baselines (avg/p95) for critical APIs and iterated on async execution paths to hit strict latency targets for customer-facing dashboards.
- Optimized backend APIs via async processing and concurrency, improving response times and throughput under load.

Projects

EarningsLens (Python, Gradio, Hugging Face, Reinforcement Learning) Live Demo [🔗](#) | GitHub [🔗](#) | Dataset [🔗](#)

- Built an RL environment to predict post-earnings stock movement using earnings call transcripts and official press releases.
- Curated and published an open-source dataset to support reproducible experimentation and model benchmarking.

ChatGPT Tokenizer (NextJS, ReactJS, TypeScript) Live Demo [🔗](#) | GitHub [🔗](#)

- Manual implementation of the Byte Pair Encoding (BPE) algorithm used by OpenAI in GPT-4.

NotebookLM (NextJS, TypeScript, LangChain, Pinecone, PostgreSQL) Demo Video [🔗](#) | GitHub [🔗](#)

- Built a RAG chatbot for accurate, domain-specific question answering.
- Developed end-to-end document ingestion and retrieval workflows for long-form content.

Kanban App (React.js, Node.js, Express.js, MongoDB) GitHub: [Front End 🔗](#) | [Back End 🔗](#)

- Built a full-stack Kanban application from scratch.
- Added authentication, task management, and end-to-end frontend-backend integration.

Skills

Programming Languages: Python, TypeScript, JavaScript, Go

Back End: FastAPI, Flask, ExpressJS, Node.js, REST APIs, GraphQL, Microservices, Gin

Front End: Next.js, React.js, Tailwind CSS, Redux Toolkit, Zustand, HTML/CSS

AI & LLMs: RAG, LangChain, Prompt Engineering, Vector Databases, Pinecone, Langfuse

Data & Messaging: SQL, NoSQL, PostgreSQL, MongoDB, Redis, Firebase, RabbitMQ, Database Design

Cloud & DevOps: AWS, GCP, Docker, NGINX, CI/CD Pipelines, Linux, Observability, Monitoring, Logging